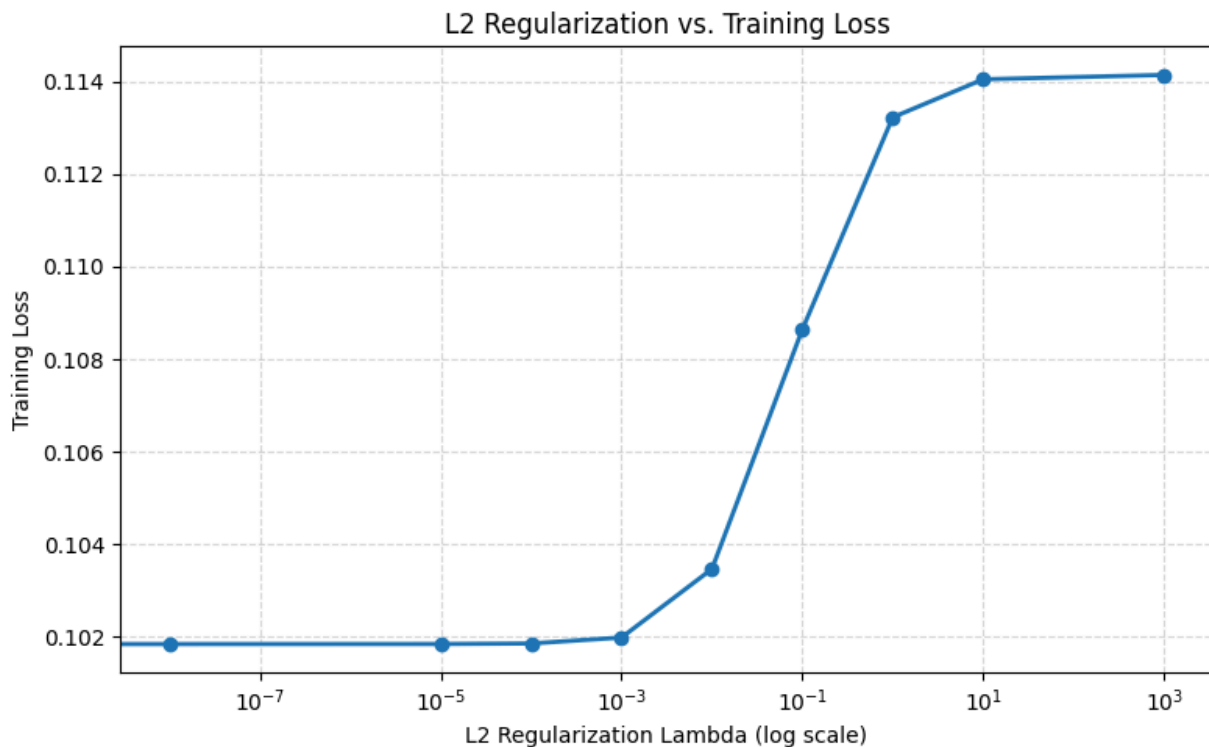


L2 regularization and training loss:



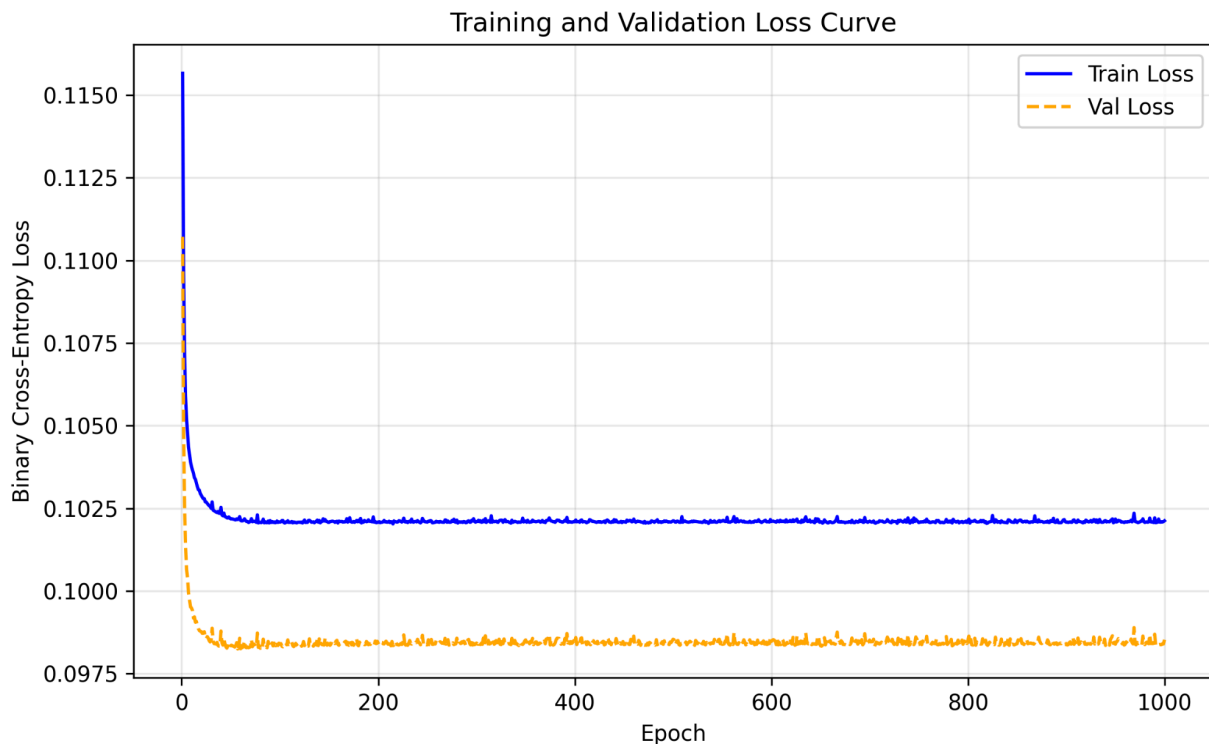
I generated this graph using my final set of features (and not the age-only model) because I saw the requirement for this plot too far into the project and didn't feel like running everything again. The main trends of how values of L2 regularization lambda affect training loss are still visible. At values lower than 10^{-3} , there is minimal reduction in training loss for progressively larger lambdas. This is because lambda is so small at that point that it is essentially 0, and therefore has no impact on the loss function. On the opposite end of the spectrum, however, lambda is too big and outweighs any weights. This is why it also plateaus at large lambda values because the actual weights no longer matter. Additionally, I can tell you from my mistakes that lambdas greater than 10^3 will cause overflow errors.

Part 2

I used the following features and parameters:

```
feature_config = {  
    "numerical": ["age", 'pack_years', 'smokea_f'],  
    "categorical": ["sex", "race7", 'bronchit_f', 'lung_fh'],  
    "ordinal": ["educat", 'bmi_50c']  
}  
  
command.extend(["--learning_rate", '0.01'])  
command.extend(["--batch_size", '64'])  
command.extend(["--num_epochs", '1000'])  
command.extend(["--regularization_lambda", '0.01'])
```

2.1: Ablation Study

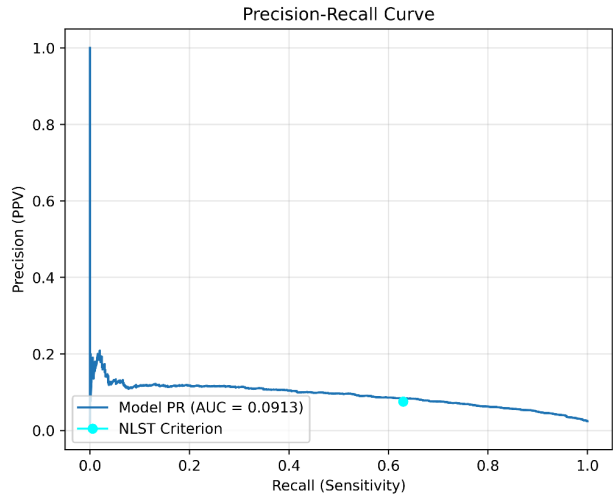
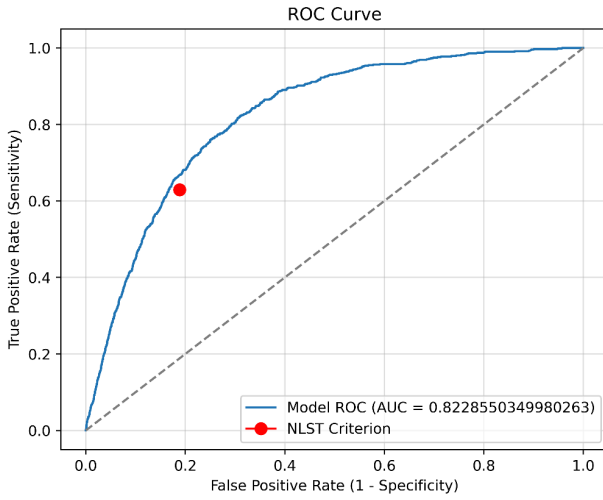


The loss curves decrease rapidly in the first ~20 epochs before plateauing to steady loss values for both training loss and validation loss. After about 100 epochs, the model got as good as it will get. I used a high epoch to allow large regularization lambda values to converge for 1.2. The minimal gap between training and validation loss curves shows that the model isn't overfitting while still being able to generalize. Also, since we see minimal oscillations in the loss curves, that means the learning rate and batch size parameters I chose are stable steps in SGD.

For achieving good performance, the clipping that we learned in lecture was very helpful. I actually implemented it in the wrong step in logistic regression at first, so I thought it wasn't

doing anything at first. Another choice that helped was to normalize numerical features so that they are appropriately scaled by the model when fitting weights. Finally, I initially ran a huge grid search to fine tune my parameters overnight.

2.2: Analyzing overall model performance



Feature	Subgroup	Sample Count	Positives	AUC
sex	1.0	14888	431.0	0.8022
sex	2.0	14999	281.0	0.8425
race7	1.0	25598	603.0	0.8324
race7	2.0	1437	36.0	0.8071
race7	3.0	510	10.0	0.7604
race7	4.0	1084	26.0	0.8109
race7	5.0	179	5.0	0.8437
race7	6.0	84	1.0	0.5181
race7	7.0	995	31.0	0.5936
educat	0.0	1066	32.0	0.6037
educat	1.0	260	9.0	0.7610
educat	2.0	1803	101.0	0.7736
educat	3.0	6550	153.0	0.8152
educat	4.0	3555	92.0	0.8119
educat	5.0	6248	157.0	0.8230
educat	6.0	4971	91.0	0.8474
educat	7.0	5434	77.0	0.8414
cig_stat	0.0	14329	81.0	0.7809
cig_stat	1.0	3112	288.0	0.6554
cig_stat	2.0	12446	343.0	0.7578
nlst_flag	0.0	23943	264.0	0.7809
nlst_flag	1.0	5944	448.0	0.6510

Sensitivity: 0.0014 | Specificity: 0.9997 | Precision: 0.0909

I performed this analysis at my default threshold of 0.5. Like we talked about in class, prevalence greatly impacts how the model performs. My sensitivity is a horrible 0.0014, which means it misses almost 99.9% of positive cancer cases. Since lung cancer cases are so low, erring on the side of a negative diagnosis is vastly more likely to be true (shown by my high specificity). At this threshold, my classifier is not very useful despite having a high AUC.

My model ROC looks pretty good and is above the NLST criterion, meaning the model has a higher sensitivity at the same FPR leading to more diagnoses without false positives. My Precision-Recall curve is a little hard to read and in hindsight I should have trimmed the x axis since a recall of 0 will have close to 100% precision due to low prevalence. Starting from around 10% recall to 60%, the model has consistent PPV, showing the model maintains a steady precision across many thresholds. The PR is above the NLST criterion again showing the same conclusions as the ROC curve.

There are several AUC differences >0.05 between groups. Race is a good example. It ties together many factors such as prevalence, access to care, average socioeconomic status, and genetic predispositions to name a few. Cig_stat also has large AUC differences. This is because being a current cigarette smoker puts you at an extremely high risk of developing lung cancer. The disparity in AUC across groups shows limitations in how sparsity can rapidly increase variance in minority groups.

2.3: Model interpretation

2.3 MODEL FEATURES			
Model Intercept (Bias): -3.2780			
Rank	Feature	Weight (Theta)	Odds Ratio (e^w)
1	pack_years	0.5456	1.7256
2	age	0.2018	1.2236
3	bmi_50c	-0.1026	0.9025
4	lung_fh_1	0.0817	1.0852
5	lung_fh_0	-0.0713	0.9312

My top 3 most important features are as follows in order: pack_years, age, bmi_50c. I kept Age because it had strong performance as a single feature. Pack years logically made sense as a strong indicator of whether someone would develop lung cancer due to the vastly higher risk of cancer from regular smoking. Finally, I know obesity is highly comorbid with a plethora of diseases. I used BMI at 50 years old to track this information. I did use a continuous version of the variable but found it was less accurate than using the categorical version. I identified these features as my most important because they have the highest absolute value weight in my model.

2.4: Simulating Clinical Utility

```
2.4 SIMULATING CLINICAL UTILITY
NLST Criteria Baseline -> Sensitivity: 0.6292 | Specificity: 0.8116 | PPV: 0.0754
Matched Model          -> Sensitivity: 0.6685 | Specificity: 0.8115 | PPV: 0.0797 (Threshold: 0.0313)
SUBGROUP PERFORMANCE AT MATCHED THRESHOLD (0.0313)
Feature  Subgroup  Count  Sens  Spec  PPV
sex      1.0      14888 0.7332 0.7365 0.0766
sex      2.0      14999 0.5694 0.8852 0.0865
race7    1.0      25598 0.6766 0.8208 0.0835
race7    2.0       1437 0.5556 0.8287 0.0769
race7    3.0       510 0.5000 0.8720 0.0725
race7    4.0      1084 0.5769 0.8856 0.1103
race7    5.0       179 0.6000 0.8563 0.1071
race7    6.0        84 0.0000 0.8193 0.0000
race7    7.0       995 0.8065 0.4232 0.0430
educat   0.0      1066 0.8125 0.4333 0.0425
educat   1.0       260 0.7778 0.6335 0.0707
educat   2.0      1803 0.7426 0.6533 0.1128
educat   3.0      6550 0.6993 0.7927 0.0747
educat   4.0      3555 0.6957 0.7961 0.0831
educat   5.0      6248 0.6688 0.8194 0.0871
educat   6.0      4971 0.5934 0.8727 0.0800
educat   7.0      5434 0.4935 0.9108 0.0736
cig_stat 0.0      14329 0.3210 0.9606 0.0442
cig_stat 1.0      3112 0.7535 0.4671 0.1260
cig_stat 2.0     12446 0.6793 0.7164 0.0636
nlst_flag 0.0     23943 0.3485 0.9206 0.0467
nlst_flag 1.0      5944 0.8571 0.3415 0.0959
```

1. Sensitivity, specificity, and PPV of NLST criteria on PLCO test set are calculated in NLST Criteria Baseline.
2. If I match the NLST's specificity, my model achieves a sensitivity of 0.6685 (an improvement over NLST).
3. At a high threshold, the model is essentially worthless as a diagnostic tool. I matched the model's specificity to the NLST's and the p value for that specificity was 0.0313. Maintaining the same specificity means that there is the same number of true negatives out of all negative decisions. So my model does not introduce more (proportionally) false negatives into the mix. However, at the same specificity, my model has a higher sensitivity, meaning that more people who would have been incorrectly diagnosed as healthy will instead get the care they need.
4. Metrics for each subgroup at my threshold of 0.0313 are in the image.

2.5: Identifying limitations in study design

This dataset is collected from Prostate, Lung, Colorectal, and Ovarian (PLCO) exams. Participants in these studies are not necessarily representative of the general US population.

For example, they might be better educated or better able to afford healthcare services and are therefore healthier. This difference between sample and population can also affect many other factors such as smoking prevalence. Also, we can see a disparity in how many minorities are screened. While the US population does not have an equal distribution of ethnicities, having one ethnic population sampled much more heavily than another leads to biases within the model. One easy additional study that could be done to better inform screening guidelines would be to test this model on various datasets from the real world that cover many different populations. Another study that would be helpful would be an analysis of the real world implications of using this model. While there is a concrete improvement in sensitivity, it is not so clear what that percentage increase means in real world impacts. We could measure financial costs and clinical impact from using this tool.